NOT MEASUREMENT SENSITIVE

A-A-59168B
9 July 2018
SUPERSEDING
A-A-59168A
21 May 2014

COMMERCIAL ITEM DESCRIPTION
LAYOUT FLUID (DYE), BLUE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-L-83795B for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for a blue layout fluid (dye) used to coat metal and glass surfaces for scribing or marking purposes.

2. CLASSIFICATION. The blue layout fluid shall be classified into three products container Types.

Type 1	Aerosol container
Type 2	Applicator-in-cap container
Type 3	Screw-top container

3. SALIENT CHARACTERISTICS.

3.1 Materials.

3.1.1 Fluid composition. The blue layout fluid shall be a uniform mixture of suitable organic solvents, film former, and blue dye with a sweet solvent odor. The mixture shall be free of sediments and suspended particles that can be detected by visual examination.

3.1.2 Propellant. The propellant for Type 1 products shall consist of a mixture of propane and isobutane.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DLA Aviation, VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

3.2 Physical properties.

3.2.1 Color. The layout fluid shall be blue such that it will match or be darker than Color 25102 within SAE AMS-STD-595, Colors Used in Government Procurement, by visual comparison.

3.2.2 Flash point. The layout fluid shall have a minimum flash point of 53ºF (11.7ºC) when tested in accordance with ASTM D56, Standard Test Method for Flash Point by Tag Closed Cup Tester. For Type 1 products, this requirement shall be verified on the bulk layout fluid prior to aerosol packaging.

3.2.3 Viscosity. The layout fluid shall have a maximum efflux time of 32 seconds when tested using Gardco Zahn Signature Series viscosity cup No.1. The manufacturer's instructions for the use and care of the Zahn cup shall be followed.

3.2.4 Dry film. A layer of layout fluid applied onto a dry, smooth, and flat ferrous metal surface in accordance with ASTM D823, Standard Practices for Producing Films of Uniform Thickness of Paint, Coatings and Related Products on Test Panels, Practice E, shall have the following properties:

3.2.4.1 Drying time. The film shall air-dry within no more than five minutes such that the coating will not visibly rub off on finger by light rubbing.

3.2.4.2 Appearance. The dry film shall be visually non-glaring with a uniform finish and color.

3.2.4.3 Thickness. The dry film shall have a maximum thickness of 0.3 mils (0.0003 inches) when measured in accordance with ASTM D7091, Standard Practice for Nondestructive Measurement of Dry Film Thickness of Nonmagnetic Coatings Applied to Ferrous Metals and Nonmagnetic, Nonconductive Coatings Applied to Non- Ferrous Metals.

3.2.4.4 Adhesion. The dry film shall not flake or peel such that when tested in accordance with ASTM D3359, Standard Test Methods for Rating Adhesion by Tape Test, Test Method B, shall have a classification rating of 5.

3.3 Chemical resistance. The layout fluid dried on a metal or glass surface shall not wash away with synthetic coolants, chlorinated solvents, oils, cutting oils, or water. The manufacturer shall provide safe removal instructions.

3.4 Containment.

3.4.1 Pressurized container. Pressurized container of Type 1 products shall not leak, and shall contain 25 to 30 percent by weight of the required propellant to completely empty the fluid contents over a normal period of use.

3.4.2 Non-pressurized container. Type 3 products container shall have screw-top closures and Type 2 products container shall incorporate an applicator such as a brush or a dauber, as an integral part of the closure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

4. REGULATORY REQUIREMENTS.

4.1 Safety data sheet (SDS). Manufacturers shall prepare and submit an SDS in accordance with FED-STD-313, Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities, and the Code of Federal Regulations (29 CFR 1910.1200).

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The blue layout fluid products provided shall meet the salient characteristics requirements of this commercial item description; shall conform to the producer's own drawings, specifications, standards, and quality assurance practices; and shall be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market acceptability criteria. The blue layout fluid products procured under this commercial item description shall have been previously sold to the government or commercial customers.

6. PACKAGING.

6.1 Preservation, packing, and marking. The blue layout fluid products shall be preserved, packed, and marked as specified in the acquisition contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 ASTM Standards may be obtained online at https://www.astm.org/.

7.1.2 The CFR may be obtained online at https://www.ecfr.gov/.

7.1.3 FED-STD-313 is available online at http://quicksearch.dla.mil/.

7.1.4 SAE Standards may be obtained online at https://www.sae.org/.

7.2 National stock number (NSN). The following assigned NSNs correspond to this CID.

6850-01-015-0834 (Type 1, 12 ounces net weight)
6850-00-584-2879 (Type 2, 4 ounces, quantity of 12 total)
6850-00-985-6227 (Type 2, 8 ounces)
6850-00-664-9067 (Type 3, 1 pint)
6850-01-276-5396 (Type 3, 1 gallon)

Other NSNs may also correspond to the blue layout fluid specified in this commercial item description.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

7.3 Part numbers. For cataloging purposes, part numbers associated with this commercial item description may be A-A-59168-* with the final character (*) being 1, 2, or 3 to designate the Type of container (see 2).

MILITARY INTERESTS:

Custodians:
Navy – AS
Air Force – 68
DLA – GS

Reviewer:
Navy – SH

Civil Agency
Coordinating Activity
GSA – FAS

Preparing activity:
DLA – GS3

(Project 6850-2018-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.